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ENG 687 Assignment: Aerodynamic Investigation into Pressure Distribution and Coefficients

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S23007133 Aerodynamics

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ENG 687 Assignment: Aerodynamic Investigation into Pressure Distribution and Coefficients

Introduction

In the field of engineering, understanding the behavior of fluid flow around objects is fundamental for optimizing design and performance. This study investigates two essential cases in aerodynamics: the pressure distribution around a circular cylinder and the lift and drag characteristics of a NACA 0012 aerofoil. These analyses are critical for applications ranging from aircraft wings to automotive designs.

Objectives:

1. To analyze and compare theoretical and experimental surface pressure distributions on a circular cylinder.
2. To calculate and interpret lift and drag coefficients for a 3D NACA 0012 aerofoil, emphasizing its relevance in aerodynamic wing design.

A circular cylinder immersed in uniform flow experiences pressure differences due to fluid interaction. These pressure variations lead to a drag force, primarily caused by form drag, resulting from the pressure differential between the front and rear surfaces of the cylinder.

Figure 1 illustrates the streamline patterns around a cylinder when subjected to a uniform flow. The higher pressure on the front half compared to the rear half results in a net drag force.

Figure 1: Streamlines around a circular cylinder in uniform flow.

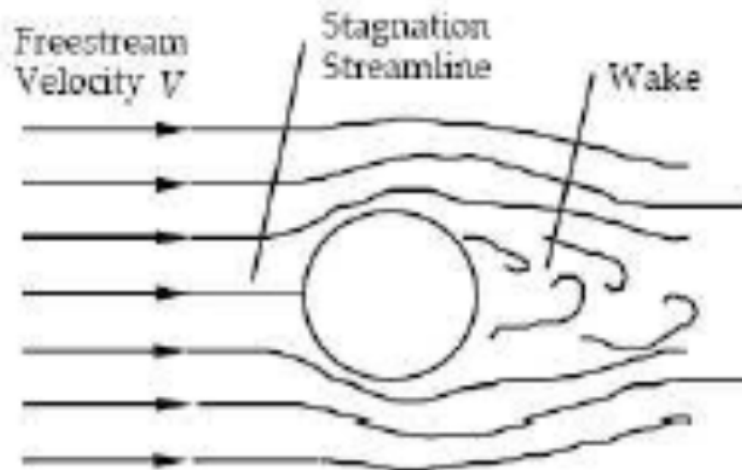


Figure 1 demonstrates how the uniform flow interacts with a circular cylinder, creating distinct patterns of streamlines. The stagnation streamline splits the flow, with some streamlines moving over the top and others beneath the cylinder. At the stagnation point, located at the front of the cylinder, the fluid velocity drops to zero, resulting in maximum pressure. As the fluid continues around the cylinder, a pressure gradient forms, leading to a wake region at the rear where the flow separates from the surface. This wake region is characterized by low-pressure zones, which contribute significantly to the form drag acting on the cylinder. Understanding these flow patterns is essential for analysing pressure-induced drag and optimizing designs to minimize energy losses.

The pressure distribution on the cylinder surface can be visualized in Figure 2. Arrows indicate the pressure magnitude and direction at various points. As shown, the rear half of the cylinder experiences significantly lower pressure due to flow separation and the formation of a wake region, which is critical in understanding drag behaviour.

Figure 2: Surface pressure distribution on a circular cylinder.

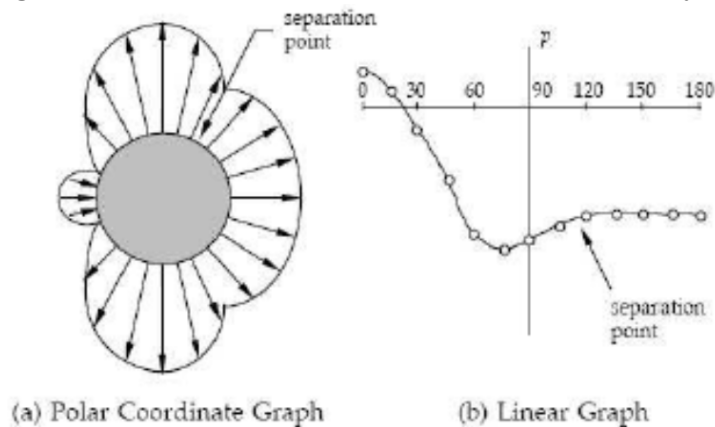


Figure 2: Surface Pressure Distribution on a Circular Cylinder

Figure 2 illustrates the surface pressure distribution around a circular cylinder. The diagram on the left (a) uses polar coordinates to represent pressure values, with the arrow lengths indicating pressure magnitudes at specific points. The separation point, where the flow detaches from the surface, is clearly marked. On the right (b), a linear graph shows the pressure coefficient, C_p , as a function of the angular position, θ . This plot highlights the steep drop in pressure on the rear half due to flow separation and wake formation.

The combination of these visualizations underscores the role of pressure gradients in creating drag forces, termed form drag, and their significance in engineering design. On the front half of the cylinder, pressures are higher due to the direct impingement of the flow, while on the rear half, flow separation leads to a significant drop in pressure. This difference in pressure between the front and rear halves is the primary contributor to the drag force on the cylinder. Understanding this distribution is essential for analysing how pressure variations result in energy losses and how these effects can be mitigated in engineering applications.

1. **Theoretical Pressure Coefficient (C_p):**

$$C_p = 1 - 4 \sin^2 \theta \quad C_p = 1 - 4 \sin^2 \theta$$

- This equation predicts the ideal pressure distribution around a circular cylinder under inviscid flow conditions. It assumes no separation and a symmetrical profile between the front and rear sides of the cylinder.

2. Experimental Pressure Coefficient (Cp):

$$C_p = \frac{p - p_\infty}{0.5\rho V^2} \quad C_p = \frac{p - p_\infty}{0.5\rho V^2}$$

- Where:
 - p : Measured surface pressure.
 - p_∞ : Free-stream pressure.
 - ρ : Fluid density.
 - V : Free-stream velocity.
- This equation accounts for boundary layer effects and wake formation, highlighting discrepancies from the idealized case.

Equations for Figure 3:

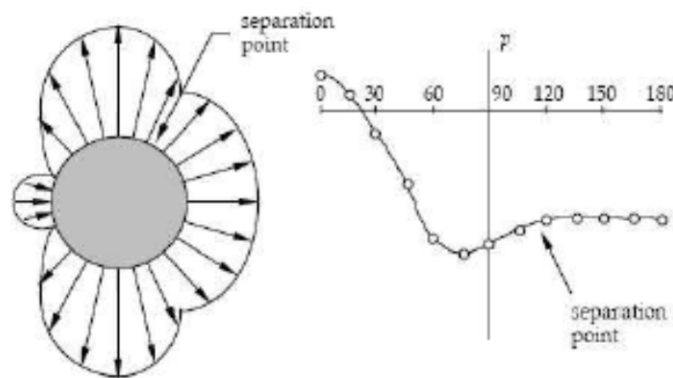


Figure 3. (a) Polar Coordinate Graph (b) Linear Graph

The discrepancies between theoretical and experimental pressure distributions, as shown in Figure 3, highlight the limitations of inviscid flow assumptions. The experimental results account for boundary layer separation, leading to a larger pressure drag on the cylinder's rear side. This difference explains why drag persists even at very low viscosities, an observation central to D'Alembert's paradox.

Equations for Figure 3:

1. Theoretical Pressure Coefficient (Cp):

$$C_p = 1 - 4\sin^2\theta \quad C_p = 1 - 4\sin^2\theta$$

- Theoretical predictions assume inviscid, irrotational flow, leading to symmetrical pressure distributions.

2. Experimental Pressure Coefficient (Cp):

$$C_p = \frac{p - p_\infty}{0.5\rho V^2} \quad C_p = \frac{p - p_\infty}{0.5\rho V^2}$$

- The experimental curve deviates significantly from theoretical values due to boundary layer separation and wake formation, which increase pressure drag on the cylinder's rear side.

Figure 3: Comparison of Theoretical and Experimental Pressure Distributions on a Circular Cylinder

Figure 3 highlights the distinctions between theoretical predictions and experimental observations of pressure distribution around a circular cylinder. The theoretical curve assumes inviscid flow, where no separation occurs, leading to symmetric pressure profiles between the front and rear halves. In contrast, the experimental curve accounts for boundary layer effects, revealing significant pressure deficits on the rear half due to flow separation and wake formation. These differences underscore the importance of considering viscous effects in practical applications.

The analysis of these profiles provides insight into the limitations of idealized models and the necessity of experimental validation for accurate aerodynamic predictions. Understanding these distinctions is essential for identifying the impact of boundary layer separation, which results in drag due to wake formation.

In the case of aerofoils, the flow dynamics also vary significantly with the angle of attack. At a zero-degree angle, as shown in Figure 4a, the flow remains attached to the aerofoil, resulting in smooth pressure variations across the surface. However, at a 5-degree angle of attack (Figure 4b), an adverse pressure gradient develops on the rear side of the aerofoil, causing boundary layer separation. This significantly impacts lift and drag forces, reducing aerodynamic efficiency. Understanding these variations is crucial for optimizing aerofoil performance, as the interplay of favourable and adverse pressure gradients dictates the onset of flow separation.

This comparison underscores the importance of experimental validation to address the limitations of idealized models, ensuring accurate predictions of aerodynamic behaviour in real-world applications.

In the case of aerofoils, the flow dynamics vary with the angle of attack. At a zero-degree angle, as shown in Figure 4a, the flow remains attached, with smooth pressure variations across the surface. However, at a 5-degree angle of attack (Figure 4b), an adverse pressure gradient on the rear side can cause boundary layer separation, significantly affecting lift and drag forces. Understanding these variations is crucial for optimizing aerofoil performance.

Figure 4a: Flow over an aerofoil at zero angle of attack.

Figure 4b: Flow over an aerofoil at a 5-degree angle of attack.

Flow dynamics for aerofoils vary significantly with the angle of attack, as depicted in Figures 4a and 4b. In Figure 4a, where the angle of attack is zero degrees, the flow remains largely attached to the aerofoil surface, creating smooth and uniform pressure variations. This condition minimizes drag and allows for efficient aerodynamic performance. Conversely, Figure 4b shows the scenario at a 5-degree angle of attack, where an adverse pressure gradient develops on the rear side. This adverse gradient can lead to boundary layer separation, significantly increasing drag and reducing lift. Such variations are crucial for understanding aerofoil performance, especially in optimizing designs for specific flight conditions.

Through wind tunnel experiments, this study integrates theoretical predictions and experimental observations, providing insights into fluid flow behaviour. These findings aim to enhance the understanding of drag reduction strategies and improve the aerodynamic efficiency of various designs.



RESULT, ANALYSIS AND DISCUSSION

Part 1: Pressure Distribution on a Circular Cylinder

The study of pressure distribution on a circular cylinder provides valuable insights into aerodynamic drag. Using Bernoulli's principle:

$$p + \frac{1}{2}\rho v^2 + \rho gh = \text{constant} \quad p + \frac{1}{2}\rho v^2 + \rho gh = \text{constant} \quad p + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$$

We can relate velocity and pressure changes around the cylinder. Static and dynamic pressure combine to produce the observed pressure variations, which can be expressed using the pressure coefficient equations:

1. Theoretical Pressure Coefficient:

$$C_p = 1 - 4\sin^2(\theta) \quad C_p = 1 - 4\sin^2(\theta) \quad C_p = 1 - 4\sin^2(\theta)$$

2. Experimental Pressure Coefficient:

$$C_p = \frac{p - p_\infty}{0.5\rho V^2} \quad C_p = \frac{p - p_\infty}{0.5\rho V^2} \quad C_p = 0.5\rho V^2 p - p_\infty$$

where:

- p : Measured surface pressure (Pa)
- $p_\infty = 101300 \text{ Pa}$: Free-stream pressure
- $\rho = 1.20 \text{ kg/m}^3$: Air density
- $V = 32.31 \text{ m/s}$: Free-stream velocity
- $0.5\rho V^2 = 627.53 \text{ Pa}$: Dynamic pressure

Theoretical Calculations

Using the formula for theoretical C_p

, the calculations for each angular position (θ) are as follows:

- At $\theta = 0^\circ$:
 $C_p = 1 - 4\sin^2(0^\circ) = 1.00$
- At $\theta = 30^\circ$:
 $C_p = 1 - 4\sin^2(30^\circ) = 0.00$
- At $\theta = 60^\circ$:
 $C_p = 1 - 4\sin^2(60^\circ) = -2.00$
- At $\theta = 90^\circ$:
 $C_p = 1 - 4\sin^2(90^\circ) = -3.00$
- At $\theta = 120^\circ$:
 $C_p = 1 - 4\sin^2(120^\circ) = -2.00$
- At $\theta = 150^\circ$:
 $C_p = 1 - 4\sin^2(150^\circ) = 0.00$

Experimental Calculations

Using the experimental surface pressure (ppp) values provided, the C_p values were calculated as follows:

$$At \theta = 0^\circ \theta = 0^\circ \theta = 0^\circ (p = 62323 p = 62323 p = 62323):$$

$$C_p = \frac{62323 - 101300}{627.53} = -61.88$$

$$At \theta = 30^\circ \theta = 30^\circ \theta = 30^\circ (p = 61408 p = 61408 p = 61408):$$

$$C_p = \frac{61408 - 101300}{627.53} = -63.25$$

$$At \theta = 60^\circ \theta = 60^\circ \theta = 60^\circ (p = 62846 p = 62846 p = 62846):$$

$$C_p = \frac{62846 - 101300}{627.53} = -61.33$$

$$At \theta = 90^\circ \theta = 90^\circ \theta = 90^\circ (p = 61208 p = 61208 p = 61208):$$

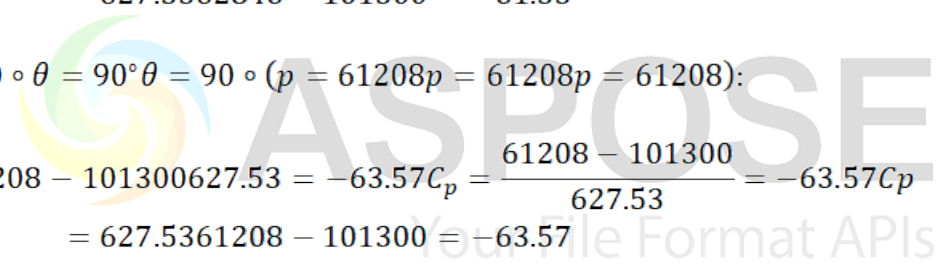
$$C_p = \frac{61208 - 101300}{627.53} = -63.57$$

$$At \theta = 120^\circ \theta = 120^\circ \theta = 120^\circ (p = 58946 p = 58946 p = 58946):$$

$$C_p = \frac{58946 - 101300}{627.53} = -66.05$$

$$At \theta = 150^\circ \theta = 150^\circ \theta = 150^\circ (p = 61431 p = 61431 p = 61431):$$

$$C_p = \frac{61431 - 101300}{627.53} = -63.21$$



Angular Position (°)	Surface Pressure (Pa)	Experimental Cp	Theoretical Cp
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TABLE OF RESULT

0	62323	$\begin{aligned} 62323 - 101300627.53 &= -61.88 \frac{62323 - 101300}{627.53} = -61.88627.5362323 - 101300 \\ &= -61.88 \end{aligned}$	$1 - 4 \sin^2(0^\circ) = 1.001 - 4 \sin^2(0^\circ) = 1.001 - 4 \sin^2(0^\circ) = 1.00$
30	61408	$\begin{aligned} 61408 - 101300627.53 &= -63.25 \frac{61408 - 101300}{627.53} = -63.25627.5361408 - 101300 \\ &= -63.25 \end{aligned}$	$1 - 4 \sin^2(30^\circ) = 0.001 - 4 \sin^2(30^\circ) = 0.001 - 4 \sin^2(30^\circ) = 0.00$
60	62846	$\begin{aligned} 62846 - 101300627.53 &= -61.33 \frac{62846 - 101300}{627.53} = -61.33627.5362846 - 101300 \\ &= -61.33 \end{aligned}$	$1 - 4 \sin^2(60^\circ) = -2.001 - 4 \sin^2(60^\circ) = -2.001 - 4 \sin^2(60^\circ) = -2.00$
90	61208	$\begin{aligned} 61208 - 101300627.53 &= -63.57 \frac{61208 - 101300}{627.53} = -63.57627.5361208 - 101300 \\ &= -63.57 \end{aligned}$	$1 - 4 \sin^2(90^\circ) = -3.001 - 4 \sin^2(90^\circ) = -3.001 - 4 \sin^2(90^\circ) = -3.00$
120	58946	$\begin{aligned} 58946 - 101300627.53 &= -66.05 \frac{58946 - 101300}{627.53} = -66.05627.5358946 - 101300 \\ &= -66.05 \end{aligned}$	$1 - 4 \sin^2(120^\circ) = -2.001 - 4 \sin^2(120^\circ) = -2.001 - 4 \sin^2(120^\circ) = -2.00$
150	61431	$\begin{aligned} 61431 - 101300627.53 &= -63.21 \frac{61431 - 101300}{627.53} = -63.21627.5361431 - 101300 \\ &= -63.21 \end{aligned}$	$1 - 4 \sin^2(150^\circ) = 0.001 - 4 \sin^2(150^\circ) = 0.001 - 4 \sin^2(150^\circ) = 0.00$



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EQUATION USED

Dynamic Pressure:

$$\begin{aligned} q &= 12 \rho V^2 = 12 \times 1.20 \times (32.31)^2 = 627.53 \text{ Pa} \\ q &= \frac{1}{2} \rho V^2 = \frac{1}{2} \times 1.20 \times (32.31)^2 \\ &= 627.53 \text{ Pa} \\ q &= 21 \rho V^2 = 21 \times 1.20 \times (32.31)^2 = 627.53 \text{ Pa} \end{aligned}$$

Experimental Cp:

$$\begin{aligned} C_p &= \frac{p - p_\infty}{q} = \frac{\text{Surface Pressure} - \text{Free-stream Pressure}}{\text{Dynamic Pressure}} \\ C_p &= \frac{p - p_\infty}{q} = \frac{\text{Surface Pressure} - \text{Free-stream Pressure}}{\text{Dynamic Pressure}} \\ C_p &= \frac{p - p_\infty}{q} = \frac{\text{Surface Pressure} - \text{Free-stream Pressure}}{\text{Dynamic Pressure}} \end{aligned}$$

Theoretical Cp:

$$C_p = 1 - 4 \sin^2(\theta) \quad C_p = 1 - 4 \sin^2(\theta) \quad C_p = 1 - 4 \sin^2(\theta)$$



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Part 2: Lift and Drag Coefficients for NACA 0012 Aerofoil

Experimental Setup

The wind tunnel experiment was conducted to measure the aerodynamic performance of the NACA 0012 aerofoil. The setup included:

- Equipment: Wind tunnel, force sensors, and NACA 0012 aerofoil.
- Conditions
: Uniform air flow at a velocity of 32.31 m/s
/with angles of attack (α) ranging from -2° to 20°
• Lift force (N),
- Reference Area (A):
The projected area of the aerofoil was calculated as: $A = \text{chord length} \times \text{span}$
 $= 0.15 \text{ m} \times 0.3 \text{ m} = 0.045 \text{ m}^2$. $A = \text{chord length} \times \text{span} = 0.15 \text{ m} \times 0.3 \text{ m}$
 $= 0.045 \text{ m}^2$. $A = \text{chord length} \times \text{span} = 0.15 \text{ m} \times 0.3 \text{ m} = 0.045 \text{ m}^2$.

Equations Used:

$$\text{Lift Coefficient (CL)}: CL = \frac{L}{0.5 \rho V^2 A} \quad CL = 0.5 \rho V^2 A L$$

$$\text{DragCoefficient}(CD): CD = \frac{D}{0.5\rho V^2 A} = \frac{D}{0.5\rho V^2 A} \quad CD = 0.5\rho V^2 A$$

$$\text{Lift-to-DragRatio}(L/D): L/D = \frac{C_L}{C_D} \quad L/D = \frac{C_L}{C_D}$$

○ Where:

- L : Lift force (N),
- D : Drag force (N),
- $\rho = 1.20 \text{ kg/m}^3$: Air density,
- $V = 32.31 \text{ m/s}$: Free-stream velocity,
- $A = 0.045 \text{ m}^2$: Reference area.

Results

The calculated lift and drag coefficients, along with the lift-to-drag ratios, are presented in Table 1.

Table 1: Aerodynamic Coefficients for NACA 0012 Aerofoil

Angle of Attack (α)	Lift Force (N)	Drag Force (N)	C_L	C_D	L/D
-2	-2.29				

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